

|   |     |       |                                                                                                                                                                                                                                                                                                                                                        |          |
|---|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 5 | (a) | (i)   | The magnetic flux through a plane surface is the product of the flux density normal to the surface and the area of the surface.                                                                                                                                                                                                                        | B1       |
|   |     | (ii)  | $\phi_A = \phi_B$<br><br>As the magnetic flux (field lines) are confined within the iron core, the magnetic flux that passes through coils A, B and C are the same.                                                                                                                                                                                    | B1       |
|   |     | (iii) | <u>Coil A, connected to AC source, produces a magnetic flux density that is varying with time sinusoidally. This results in a varying magnetic flux linkage through coil A and hence a varying magnetic linkage through coil B.</u><br><br><u>By Faraday's law, an emf is induced in coil B due to changing magnetic flux linkages through coil B.</u> | M1<br>A1 |
|   |     | (iv)  | $\frac{N_A}{N_B} = \frac{V_A}{V_B}$ $\frac{11}{6} = \frac{240}{V_B}$ $V_B = 131 \text{ V}$                                                                                                                                                                                                                                                             | M1<br>A1 |
|   | (b) |       | Since soft iron has a gap, the <u>magnetic flux through coil C will be reduced and hence the rate of change of magnetic flux linkage through coil C will be lowered.</u><br><br>By Faraday's law, <u>induced emf in coil C is proportional to the rate of change of magnetic flux linkages</u> , hence the <u>induced emf will be lower.</u>           | M1<br>A1 |
|   | (c) |       |                                                                                                                                                                                                                                                                                                                                                        |          |

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|---|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------|--|
|   |     | <p>[M1] for correct shape (positive cosine graph can only be in either the positive or negative axis).</p> <p>[A1] for correct label of <math>\varepsilon_0</math> and <math>T/4</math>, <math>3T/4</math>, <math>5T/4</math>, <math>7T/4</math> (no values required).</p> |   |                                                                    |  |
| 6 | (a) | (i)                                                                                                                                                                                                                                                                        | 1 | $6 \times (10^9) \times 10^9 \times 4 \times 10^9 \times 10^{-19}$ |  |